

K.J. Somaiya Institute of Technology
Department of Artificial Intelligence and Data Science
Project-Based Learning (PBL) – One Page Project Summary

Academic Year : 2026-27

Semester : 3

1. PROJECT DATA

Project Title: Pulse Price Volatility Analyzer

Project Domain: Agricultural Market Monitoring / Data Structures & Algorithms

Project Guide: Pradnya Jadhav

Team Members: Thanki Aditya Harish (Roll No.48), Darsh Ved (Roll No.55),
[Name 3] (Roll No.__)

2. EXECUTIVE SUMMARY

Pulse Price Volatility Analyzer processes historical grain wholesale price records to detect and flag abnormal fluctuations. Using Java sorting and searching algorithms, the system computes price changes, ranks them, and highlights records that exceed a configurable threshold. The prototype is lightweight, systematic, and suitable for both academic demonstration and practical market monitoring.

3. PROBLEM STATEMENT

Wholesale grain price datasets are large and complex, making manual inspection slow and error-prone. The project addresses this gap by automating detection of unusually large price movements, enabling faster and more reliable market analysis.

4. KEY OBJECTIVES

- Ingest and validate historical grain price records.
- Compute absolute and percentage price changes.
- Implement sorting and searching for efficient record retrieval.

- Flag abnormal fluctuations based on configurable thresholds.

5. METHODOLOGY & TECH STACK

Algorithms/Models: Price-change calculation; Sorting (Java Collections sort, selection/insertion for demo); Searching (binary/linear); Threshold-based flagging.

Tools & Frameworks: Java 11+, Maven/Gradle, OpenCSV/Apache Commons CSV, JUnit.

Dataset Used: Agricultural wholesale grain price datasets (e.g., Agmarknet, APMC mandi data); synthetic datasets with injected anomalies.

Hardware/Environment: Standard laptop with JVM; optional local database for larger datasets.

6. OUTCOMES & DELIVERABLES

Working Prototype/Model: Java application that loads CSVs, computes changes, sorts, searches, and flags anomalies.

Code Repository: <https://github.com/Darsh211107/pulse-price-volatility-analyzer.git>

Key Findings: Sorting and searching reduce analyst effort; thresholds control false positives; simple Java implementation demonstrates core algorithm concepts in a real-world context.

Project Guide Name and sign: